Dynamic Programming
+
Divide and Conquer
Sheet

Question 1: Mars Rover Debris Navigation

Problem Description:
A remote-controlled Mars Rover lands on a Martian grid of size M x N, where M denotes the total number of rows and N denotes the total number of columns. The rover starts at the drop zone at the top-left cell (0, 0) and must reach the extraction zone at the bottom-right cell (M-1, N-1). The rover is programmed to only move South(down) or East(right) by exactly one cell at a time. Furthermore, a recent storm has scattered debris across the sector. In the grid, a cell containing a 0 represents safe, passable terrain, while a cell containing a 1 represents impassable debris that the rover cannot enter. Calculate the total number of unique, valid paths the rover can take to reach the extraction zone.

Note:
- grid[i][j] is either 0 or 1.
- The starting cell (0, 0) and the extraction cell (M-1, N-1) are guaranteed to be 0 (free of debris).

Input Format:
The first line contains two space-separated integers, M and N.
The next M lines each contain N space-separated integers (either 0 or 1), representing the layout of the Martian grid.

Output Format:
Print an integer representing the total number of unique paths modulo $10^9 + 7$. If no valid path exists, print 0.

Sample Test Case 1:

Input:
```
3 3
0 0 0
0 1 0
0 0 0
```

Output:
```
2
```

Sample Test Case 2:

Input:
```
4 5
0 0 0 1 0
0 1 0 0 0
0 0 1 0 0
1 0 0 0 0
```

Output:
```
4
```

Question 2: The Freelance Developer's Dilemma

Problem Description:
You are a freelance software developer with exactly H hours of available working time this week. You have received a list of N distinct potential freelance projects.

For the i-th project (where $0 \leq i < N$):

- It requires exactly hours[i] to complete.
- It pays a fixed, guaranteed amount of payment[i].

Conditions:

1. **No Partial Work:** You cannot take on a fraction of a project. You must either complete the project entirely or decline it completely.
2. **No Duplicates:** You cannot take the same project more than once.
3. **Time Limit:** The total hours spent on your chosen combination of projects must be less than or equal to your total available hours, H.

Determine the maximum total payment you can earn this week without exceeding your available hours.

Input Format:
Line 1: Two space-separated integers, N (total number of projects) and H (total available hours).
Line 2: N space-separated integers representing the payment array.
Line 3: N space-separated integers representing the hours array(hours[i]≥1).

Output Format:
Print a single integer representing the maximum total payment achievable.

Sample Test Case 1:

Input:
```
3 50
60 100 120
10 20 30
```

Output:
```
220
```

Sample Test Case 2:

Input:
```
4 5
10 40 30 50
5 4 6 3
```

Output:
```
50
```

Question 3: Cryptographic Subsequence Extraction

Problem Description
You are working on a decryption algorithm. You are given two non-empty strings: a Source string and a Target string. Your task is to determine the total number of distinct ways to form the Target string from the Source string. A Target string can be formed by deleting zero or more characters from the Source string without changing the relative order of the remaining characters. Two derivations are considered distinct if they differ in the set of indices removed from the Source string.

Note
- The length of the Target string is less than or equal to the length of the Source string.
- Both strings consist only of uppercase English letters ('A' to 'Z').

Input Format
Line 1: Two space-separated integers N and M (N = length of Source string, M = length of Target string)
Line 2: A string of length N representing the Source
Line 3: A string of length M representing the Target

Output Format
Print a single integer representing the total number of distinct subsequences of the Source string that equal the Target string. If it is impossible to form the Target string, print 0.

Sample Test Case 1

Input:
```
7 6
RABBBIT
RABBIT
```

Output:
```
3
```

Sample Test Case 2
Input:
```
8 6
AAABBBCC
AABBCC
```
Output:
```
9
```

Question 4: Circular Energy

Problem Description:
You are the chief engineer of a starship with a circular power grid. Each power module has a specific integer `energy_output`.
Because the grid is circular, the first and last modules are adjacent. To prevent a catastrophic overload, no two adjacent modules can be activated simultaneously.

Given an integer array `energy_output`, return the maximum total energy you can safely generate without triggering an overload.

Input Format:
Line 1: One integer N (Number of modules)
Line 2: N integers (energy output value for each module)

Output Format:
Maximum total energy you can safely generate without triggering an overload

Sample Test Case 1:

Input:
```
4
6 2 4 5
```

Output:
```
10
```

Sample Test Case 2:

Input:
```
9
4 6 2 3 7 8 1 5 9
```

Output:
```
26
```

Question 5: Best Time to Buy and Sell Stock IV

Problem Description:
You are given an array of N integers where the ith element represents the price of a stock on day i.

You are also given an integer K representing the maximum number of transactions allowed. A transaction consists of buying and then selling one share of the stock.

Your task is to determine the maximum profit that can be achieved.

Note: You may not hold multiple stocks at once (you must sell before buying again).

Input Format:
Line 1: Two integers N and K
Line 2: N integers (stock prices)

Output Format:
Print a single integer representing the maximum profit.

Sample Test Case 1:

Input:
```
6 2
3 2 6 5 0 3
```

Output:
```
7
```

Sample Test Case 2:

Input:
```
5 1
7 1 5 3 6
```

Output:
```
5
```

Question 6: Longest Palindromic Subsequence

Problem Description:
You are given a string S of length N consisting of lowercase English letters.

Your task is to determine the length of the longest subsequence of S that forms a palindrome.

A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.

Input Format:
Line 1: One integer N
Line 2: A string S of length N

Output Format:
Print a single integer representing the length of the longest palindromic subsequence.

Sample Test Case 1:

Input:
```
5
bbbab
```

Output:
```
4
```

Sample Test Case 2:

Input:
```
7
racecar
```

Output:
```
7
```

Question 7: Wildcard Matching

Problem Description:
Given an input string (s) and a pattern (p), implement wildcard pattern matching with support for '?' and '*' where:

'?' Matches any single character.

'*' Matches any sequence of characters (including the empty sequence).

The matching should cover the entire input string (not partial).

Input Format:
Line 1: Two integers N and M (length of string S and pattern P)
Line 2: String S
Line 3: String P

Output Format:
Print YES if the string matches the pattern, otherwise print NO.

Sample Test Case 1:

Input:
```
5 3
adceb
a*b
```

Output:
```
YES
```

Sample Test Case 2:

Input:
```
2 2
cb
?a
```

Output:
```
NO
```

Question 8: Longest Bitonic Subsequence

Problem Description:
You are given an array of N integers.

A Bitonic Subsequence is a sequence that first strictly increases and then strictly decreases.

Your task is to determine the length of the longest bitonic subsequence in the given array.

Note: Either increasing or decreasing part can be empty, but the sequence must have at least one element.

Input Format:
Line 1: One integer N
Line 2: N integers (array elements)

Output Format:
Print a single integer representing the length of the longest bitonic subsequence.

Sample Test Case 1:

Input:
```
8
1 11 2 10 4 5 2 1
```

Output:
```
6
```

Sample Test Case 2:

Input:
```
5
12 11 40 5 3
```

Output:
```
5
```

Question 9: Same DNA

Problem Description:
You are a bioinformatics researcher analyzing two DNA sequences, represented as strings `SeqA` and `SeqB`.
You need to determine the minimum number of operations required to mutate `SeqA` exactly into `SeqB`. You are permitted three operations on any character in `SeqA`:

1. Insert a character
2. Delete a character
3. Replace a character

Given the two strings `SeqA` and `SeqB`, return the minimum number of operations required to achieve the mutation.

Input Format:
Line 1: Two integers N, M (length of `SeqA`, `SeqB`)
Line 2: N characters (`SeqA`)
Line 3: M characters (`SeqB`)

Output Format:
Minimum number of steps required to transform `SeqA` to `SeqB`

Sample Test Case 1:

Input:
```
4 5
ATTG
ACTTC
```

Output:
```
2
```

Sample Test Case 2:

Input:
```
7 9
ATTCGGA
TCGGATGCA
```

Output:
```
6
```

Question 10: Optimised Cuts

Problem Description:
You are a structural engineer fabricating a specialized titanium beam of overall length N.
You are given an integer array cuts representing specific positions along the beam where you must make precision cuts.
The cost of making a cut is strictly equal to the length of the beam segment being cut.
When a cut is made, the beam is permanently split into two smaller, independent segments.
You can perform the cuts in **any** order you choose.

Given the integer N and the array cuts, return the minimum total cost required to perform all the necessary cuts.

Input Format:
Line 1: Two integers N, M (Size of the beam initially, Number of cuts to be made)
Line 2: M integers (Cutting Position)

Output Format:
Minimum total cost required to perform all the necessary cuts.

Sample Test Case 1:

Input:
```
7 4
1 3 4 5
```

Output:
```
16
```

Explanation:
First cut at position 3: cost is 7. Beam splits into segments [0,3] and [3,7].
Second cut at position 5: cost is 4 (length of [3,7]). Splits into [3,5] and [5,7].
Third cut at position 1: cost is 3 (length of [0,3]). Splits into [0,1] and [1,3].
Fourth cut at position 4: cost is 2 (length of [3,5]). Splits into [3,4] and [4,5].
Final Segments: [0,1], [1,3], [3,4], [4,5], [5,7]
Total cost: 7 + 4 + 3 + 2 = 16.

Sample Test Case 2:

Input:

```
9 5
1 2 4 5 6
```

Output:

```
22
```

Explanation:

First cut at position 4: cost is 9. Beam splits into segments [0,4] and [4,9].

Second cut at position 2: cost is 4 (length of [0,4]). Splits into [0,2] and [4,9].

Third cut at position 1: cost is 2 (length of [0,2]). Splits into [0,1] and [1,2].

Fourth cut at position 6: cost is 5 (length of [4,9]). Splits into [4,6] and [6,9].

Fifth cut at position 5: cost is 2 (length of [4,6]). Splits into [4,5] and [5,6].

Final Segments: [0,1], [1,2], [2,4], [4,5], [5,6], [6,9]

Total cost: 9 + 4 + 2 + 5 + 2 = 22.

Question 11: The Unstable Reactor (Ways to Add Parentheses)

Problem Description

A nuclear reactor's pressure is computed using a mathematical expression consisting of single-digit numbers and the binary operators + (addition), - (subtraction), and * (multiplication). Due to a system failure, all parentheses in the expression were removed. Ignoring standard operator precedence, your task is to determine all possible results that can be obtained by inserting parentheses in different valid ways. To solve this efficiently, you are expected to use a divide and conquer approach.

Note:

- **Strictly Single Digits:** Each number in the expression is a single digit (0–9). Multi-digit numbers will not appear.
- **No Unary Operators:** The expression will not contain leading + or -. All operators are strictly binary.
- **Duplicates Allowed:** If multiple parenthesizations yield the same result, all occurrences must be included in the output.
- **Sorting:** Output all results in non-decreasing order, separated by spaces.
- The input expression is guaranteed to be valid.
- You must use 64-bit integers (long long in C) for all calculations.

Input Format

Line 1: An integer N representing the length of the string
Line 2: A string S of length N, representing the expression

Output Format

Print all possible computed results on a single line, separated by spaces, sorted in ascending order.

Constraints

- $1 \le N \le 19$
- The string will always start and end with a digit and contain strictly alternating digits and operators.

Sample Test Case 1

Input:

```
7
5-4*3+2
```

Output:

```
-15 -9 -5 5 5

Explanation of the 5 possible groupings:

   1. 5-(4*(3+2)) = 5-(4*5) = 5-20 = -15
```

```
2. 5-((4*3)+2) = 5-(12+2) = 5-14 = -9
3. (5-(4*3))+2 = (5-12)+2 = -7+2 = -5
4. ((5-4)*3)+2 = (1*3)+2 = 3+2 = 5
5. (5-4)*(3+2) = (1)*(5) = 5
```

Sample Test Case 2

Input:

```
9
2*3-4+5*2
```

Output:

```
-30 -24 -24 -22 -12 -8 -6 6 8 12 14 16 16 18

Explanation:
   1. 2*(3-((4+5)*2)) = 2*(3-(9*2)) = 2*(3-18) = 2*(-15) = -30
   2. 2*((3-(4+5))*2) = 2*((3-9)*2) = 2*((-6)*2) = 2*(-12) = -24
   3. (2*(3-(4+5)))*2 = (2*(3-9))*2 = (2*(-6))*2 = -12*2 = -24
   4. 2*(3-(4+(5*2))) = 2*(3-(4+10)) = 2*(3-14) = 2*(-11) = -22
   5. (2*3)-((4+5)*2) = 6-(9*2) = 6-18 = -12
   6. (2*3)-(4+(5*2)) = 6-(4+10) = 6-14 = -8
   7. ((2*3)-(4+5))*2 = (6-9)*2 = (-3)*2 = -6
   8. ((2*(3-4))+5)*2 = (-2+5)*2 = 3*2 = 6
   9. (2*(3-4))+(5*2) = (2*(-1))+10 = -2+10 = 8
   10.  ((2*3)-4)+(5*2) = (6-4)+10 = 2+10 = 12
   11.  (((2*3)-4)+5)*2 = (2+5)*2 = 7*2 = 14
   12.  2*(((3-4)+5)*2) = 2*(((-1)+5)*2) = 2*(4*2) = 2*8 = 16
   13.  (2*((3-4)+5))*2 = (2*(4))*2 = 8*2 = 16
   14.  2*((3-4)+(5*2)) = 2*((-1)+(10)) = 2*(9) = 18
```

Question 12: The Corrupted Transmission (Valid Substring Splitting)

Problem Description

A network packet containing a string of lowercase English letters was intercepted. You must extract the longest valid data block from this packet. A "valid data block" is defined as a strictly contiguous substring where every unique character present in that substring appears at least K times within that exact substring. If no such substring exists, the valid length is 0.

Input Format

Line 1: Two space-separated integers N and K, where N is the length of the string and K is the required frequency threshold
Line 2: A string S of length N, consisting only of lowercase English letters (a–z)

Output Format

Print a single integer representing the length of the longest valid substring.

Constraints

- 1 <= N <= 10^5
- 1 <= K <= 10^5
- S contains only lowercase English letters.

Sample Test Case 1

Input:
```
16 3
abcdeeefffeeeabc
```

Output:
```
9
Explanation: The largest contiguous block remaining is eeefffeee.
Inside this block, e appears 6 times and f appears 3 times. Since
both are >= 3, it is valid. Its length is 9.
```

Sample Test Case 2

Input:
```
19 4
bbaaaabcbbbaaaaabbb
```

Output:
```
11
```

Question 13: The Cybernetic Bidding War

Problem Description:
Two advanced trading algorithms, Alpha and Beta, are competing in a high-stakes auction. A sequence of N high-value data caches is laid out in a straight line. Each cache has a specific monetary value.

The rules of the auction are strict:

1. Alpha and Beta take turns. Alpha always goes first.
2. On their turn, an algorithm can only select and take exactly one data cache, and it must be from either the extreme left or the extreme right of the remaining line.
3. Both algorithms are perfectly logical, possess complete information, and play optimally. Alpha plays to maximize its own score, while Beta plays to maximize its own score. The game is zero-sum: maximizing one player's score minimizes the other's.

Calculate the maximum value Alpha can guarantee assuming both players play optimally.

Input Format:
Line 1: A single integer N, representing the number of data caches.
Line 2: N space-separated integers representing the values of the caches in order.

Output Format:
Print an integer representing the maximum total value Alpha can secure.

Sample Test Case 1:
Input:
```
4
8 15 3 7
```
Output:
```
22
Explanation: Alpha takes 7 from the right. Remaining: [8, 15, 3].
Beta takes 8, remaining:[15, 3]. Alpha takes 15. Beta takes 3.
Alpha's total = 7 + 15 = 22.
```

Sample Test Case 2:
Input:
```
6
20 30 2 2 2 10
```
Output:
```
42
Explanation: Alpha takes 10 from the right. Remaining: [20, 30, 2, 2,
2]. Beta takes 2 from the right. Remaining: [20, 30, 2, 2]. Alpha
takes 2 from the right. Remaining: [20, 30, 2]. Beta takes 20 from
the left. Remaining: [30, 2]. Alpha takes 30. Beta takes 2. Alpha's
total = 10 + 2 + 30 = 42.
```

Question 14: Finding Closest Pair

Problem Description:
You are given two disjoint sets of points, P1 and P2, on a 2-dimensional Cartesian plane. Determine the minimum Manhattan distance between any point p_i in P1 and any point p_j in P2.

Manhattan Distance between `(x1, y1), (x2, y2)` **->** `|x1 - x2| + |y1 - y2|`
The algorithm must **strictly** employ a Divide and Conquer methodology.
Expected Time Complexity: O((N + M) log(N + M))

Input Format:
Line 1: Two integers N, M (Number of points in P1 and P2, `1 ≤ N, M ≤ 5 * 10^4`)
Next N Lines: Two integers x and y(coordinates of points in P1, `-10^5 ≤ x, y ≤ 10^5`)
Next M Lines: Two integers x and y(coordinates of points in P2, `-10^5 ≤ x, y ≤ 10^5`)

Output Format:
Minimum Manhattan distance between the points.

Sample Test Case 1:

Input:
```
2 2
1 1
10 10
2 1
12 12
```

Output:
```
1
```

Sample Test Case 2:

Input:
```
4 3
-5 -5
0 0
3 4
10 -5
-6 -2
1 2
4 -5
```

Output:
```
3
```

Question 15: Max Subarray Sum with Single Mutation

Problem Description:
Given an array A of N integers, compute the maximum contiguous subarray sum.
You are permitted to square exactly one element within the chosen contiguous subarray to maximize the total sum.

The solution must utilize a Divide and Conquer approach by returning multiple states for each segment across recursive calls.

Expected Time Complexity: O(N logN)

Input Format:
Line 1: One integer N (Size of the array A, `2 ≤ N ≤ 10^4`)
Line 2: N integers (Array A elements, `-10^3 ≤ A[i] ≤ 10^3`)

Output Format:
Maximum contiguous subarray sum after squaring a single element

Sample Test Case 1:

Input:
```
6
4 2 -8 5 -6 2
```

Output:
```
75
```

Explanation: 4 + 2 + **64** + 5 = 75

Sample Test Case 2:

Input:
```
7
-9 -5 6 1 -3 4 -2
```

Output:
```
84
```

Explanation: **81** + (-5) + 6 + 1 + (-3) + 4 = 84

Question 16: Generalized Reverse Pairs

Problem Description:
Given an array A of N integers and an integer multiplier K. Determine the total number of "important reverse pairs".
An index pair (i, j) is defined as an important reverse pair if and only if:

```
i < j and A[i] > K * A[j]
```

The solution must be implemented by modifying the Merge Sort algorithm.
Standard brute-force comparisons are strictly prohibited.

Expected Time Complexity: O(N logN)

Input Format:
Line 1: Two integers N, K (Size of the array A, constant K (`2 ≤ N ≤ 10^5`)
Line 2: N integers (Array A elements, `1 ≤ A[i] ≤ 10^5`)

Output Format:
Maximum contiguous subarray sum after squaring a single element

Sample Test Case 1:

Input:
```
5 2
10 3 2 8 1
```

Output:
```
5
```

Sample Test Case 2:

Input:
```
7 3
9 2 5 8 3 1 4
```

Output:
```
4
```

Question 17: Maximum Subarray Sum (Divide and Conquer)

Problem Description:
You are given an array A of N integers (which may include negative values).

Your task is to compute the maximum possible sum of a contiguous subarray using a Divide and Conquer approach.

The solution must divide the array into two halves recursively and consider:
- Maximum subarray entirely in left half
- Maximum subarray entirely in right half
- Maximum subarray crossing the midpoint

Input Format:
Line 1: One integer N (Size of the array)
Line 2: N integers (Array A elements)

Output Format:
Print a single integer representing the maximum subarray sum.

Sample Test Case 1:

Input:
```
9
-2 1 -3 4 -1 2 1 -5 4
```

Output:
```
6
```

Sample Test Case 2:

Input:
```
5
-1 -2 -3 -4 -5
```

Output:
```
-1
```

Question 18: Count of Range Sum

Problem Description:
You are given an array A of N integers and two integers lower and upper.

Your task is to count the number of subarrays whose sum lies in the inclusive range [lower, upper].

Formally, count the number of pairs (i, j) such that:
- $0 \le i \le j < N$
- $\text{lower} \le \text{sum}(A[i...j]) \le \text{upper}$

The solution must be implemented using a Divide and Conquer approach.

Brute-force solutions will not be accepted.

Input Format:
Line 1: Three integers N, lower, upper
Line 2: N integers (Array A elements)

Output Format:
Print a single integer representing the count of valid subarrays.

Sample Test Case 1:

Input:
```
5 -2 2
-2 5 -1 3 -2
```

Output:
```
6
```

Sample Test Case 2:

Input:
```
4 0 3
1 2 -1 2
```

Output:
```
7
```

Question 19: Skyline Problem

Problem Description:
You are given N buildings, where each building is represented as three integers: left, right, height.

Each building spans from coordinate left to right (exclusive of right) and has a constant height.

Your task is to compute the skyline formed by these buildings using a Divide and Conquer approach.

The skyline is defined as a list of key points (x, height) such that:
- At each key point, the height of the skyline changes
- The output must be sorted by x-coordinate
- If multiple buildings overlap, the skyline height is the maximum height at that point

Input Format:
Line 1: One integer N
Next N lines: Three integers left, right, height

Output Format:
Print the skyline as a sequence of pairs (x height) in order.

Sample Test Case 1:

Input:
```
3
2 9 10
3 7 15
5 12 12
```

Output:
```
2 10 3 15 7 12 12 0
```

Sample Test Case 2:

Input:
```
2
1 5 11
2 4 6
```

Output:
```
1 11 5 0
```

Question 20: Minimum Cost to Merge Stones

Problem Description:
You are given an array A of N integers, where A[i] represents the number of stones in the ith pile. Your goal is to merge all these piles into exactly one pile. You are also given an integer K. In one move, you can pick exactly K consecutive piles and merge them into a single pile.
The Rules
Consecutive Only: You can only merge piles that are adjacent to each other in the current array.
Cost: The cost of a single merge operation is the sum of all stones in the K piles being merged.
Result: The K piles are removed and replaced by one new pile containing their total sum.
Goal: Find the minimum total cost to reach a single pile. If it is impossible to merge the N piles into one, return -1.

Input Format:
Line 1: Two integers N and K
Line 2: N integers (Array A elements)

Output Format:
Print a single integer representing the minimum cost to merge all piles into one. If not possible, print -1.

Sample Test Case 1:

Input:
```
4 2
3 2 4 1
```

Output:
```
20
```

Explanation:
Merge (3,2) → cost = 5 → [5,4,1]
Merge (4,1) → cost = 5 → [5,5]
Merge (5,5) → cost = 10 → total = 20

Sample Test Case 2:

Input:
```
4 3
3 2 4 1
```

Output:
```
-1
```